

Drinkable Water For Everyone

**Analysing Water Potability to Support SDG 6 Clean Water and
Sanitation & SDG 3 Good Health and Wellbeing**

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The Global Water Challenge

Clean water is the foundation of healthy societies and a sustainable planet. However, billions of people currently suffer from water scarcity and poor water quality. This crisis is rapidly accelerating, driven by population surges, rising consumption, and the impacts of climate change.

Safe drinking water is not just a development goal. It's a fundamental human right. Beyond the severe health risks caused by industrial pollution, water contamination creates massive economic burdens.

Our project directly supports this global challenge and SDG 6 and 3 by developing an AI driven solution to make water safety monitoring faster, more accessible, and more economically viable.

3 GOOD HEALTH
AND WELL-BEING



6 CLEAN WATER
AND SANITATION



Key Objectives

Finding answers to:

- Which water quality parameters have the strongest influence on water potability?
- How are different chemical properties of water related to each other?
- Are there significant differences between potable and non-potable water samples?
- Can machine learning techniques be used to accurately predict water potability?
- How can data driven approaches support sustainable water quality monitoring systems?

Understanding The Dataset

1 | pH

Evaluates the acid-base balance to determine how acidic or alkaline the water sample is.

For safe drinking water, the World Health Organization (WHO) requires this metric to fall within a neutral range of 6.5 to 8.5.

2 | Hardness

Reflects the concentration of dissolved minerals (primarily calcium and magnesium) that water absorbs as it flows through geological rock formations.

3 | Solids

Measures the total accumulation of dissolved salts, organic matter, and inorganic minerals. Highly mineralised water can negatively impact taste and appearance. The recommending is a limit between 500 and 1000 mg/L

Understanding The Dataset

4 | Chloramines

A disinfectant formed by mixing chlorine and ammonia to safely treat public water networks.

Regulatory standards consider concentrations up to 4 mg/L (ppm) safe for human consumption.

5 | Sulfate

A naturally occurring compound washed into water sources from surrounding rocks and soil.

Standard freshwater contains low amounts (3 to 30 mg/L), but specific geographical locations can have increased values.

6 | Conductivity

Indicates the water's ability to transmit an electrical current.

Since pure water is an insulator, a high electrical conductivity value indicates a high concentration of dissolved ions and solids.

Understanding The Dataset

7 | Organic Carbon

Output from decaying natural matter or synthetic pollutants.

Environmental protection agencies require levels to be kept under 2 mg/L in fully treated drinking water

8 | Trihalomethanes

Chemical byproducts that are unintentionally created when chlorine disinfectants react with organic materials already present in the water.

Levels must be monitored and kept under 80 ppm

9 | Turbidity

Assesses the cloudiness of the water caused by suspended solid particles.

Based on WHO guidelines, drinking water must remain highly transparent, staying below 5.00 NTU

Our Target



Potability

The final binary classification indicating if the water is safe for human consumption

1 = Potable/Safe,
0 = Not Potable/Unsafe

